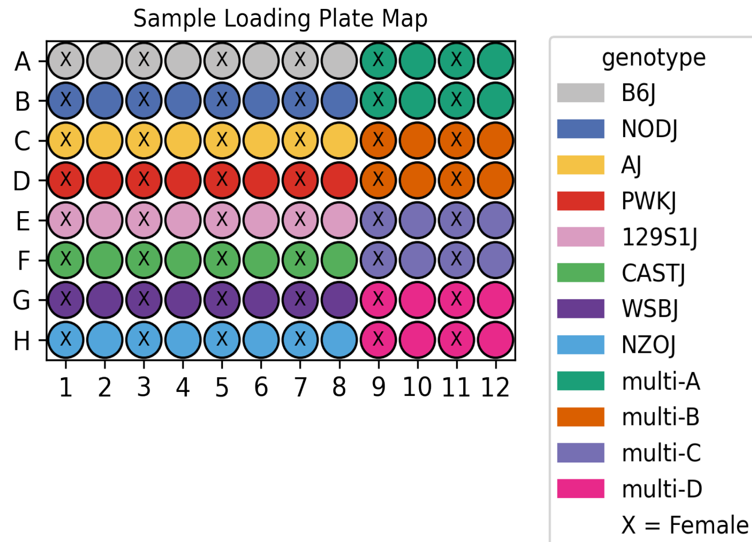
UCI/Caltech IGVF Dataset Pipeline Report

	igvf_009
Kit	WT_mega
Chemistry	Parse v2
Sample type	nuclei
Genotypes	B6J NODJ AJ PWKJ 129S1J CASTJ WSBJ NZOJ
Tissues	Adrenal Kidney
Subpools	16
Exome capture subpools	1
Expected nuclei	1,000,000



1. Sample multiplexing

Adrenal was the main tissue on the plate while Kidney was multiplexed.

Table 1: Genotype multiplexing key

Pair	Multiplexed Genotype 1	Multiplexed Genotype 2
multi-A	B6J	NODJ
multi-B	AJ	PWKJ
multi-C	129S1J	CASTJ
multi-D	NZOJ	WSBJ

2. Alignment

A total of 20,730,474,974 reads were aligned using GENCODE vM32 with mm39 assembly.

Table 2: Pseudoalignment stats

Subpool	Total reads	Total UMIs	Fraction duplicated UMIs	Total pseudoaligned	Pct. pseudoaligned
Subpool_EX	373,232,512	87,156,317	0.724	316,237,450	84.7
Subpool_2	1,474,101,220	660,068,083	0.373	1,053,123,307	71.4
Subpool_3	1,205,806,102	611,175,903	0.295	866,681,474	71.9
Subpool_4	1,400,535,478	644,666,486	0.368	1,019,370,484	72.8
Subpool_5	1,089,091,892	537,416,545	0.309	777,361,233	71.4
Subpool_6	1,645,648,594	749,156,576	0.368	1,185,773,774	72.1
Subpool_7	1,361,918,864	646,160,041	0.33	963,776,927	70.8
Subpool_8	1,321,808,185	644,354,486	0.312	936,876,581	70.9
Subpool_9	1,232,792,788	624,814,188	0.296	887,947,032	72.0
Subpool_10	1,369,324,775	646,859,161	0.343	983,866,835	71.9
Subpool_11	1,310,234,034	614,771,133	0.345	938,729,884	71.6
Subpool_12	1,181,306,371	565,158,889	0.335	850,317,858	72.0
Subpool_13	1,333,887,159	625,931,997	0.338	945,565,047	70.9
Subpool_14	1,502,670,908	686,396,547	0.366	1,082,049,533	72.0
Subpool_15	1,229,852,591	588,681,285	0.336	886,559,063	72.1
Subpool_16	1,698,263,501	417,528,430	0.653	1,202,673,009	70.8

3. Cell recovery

A total of 2,086,887 nuclei across all 16 subpools have >=200 UMIs and 927,154 nuclei have >= 500 UMIs.

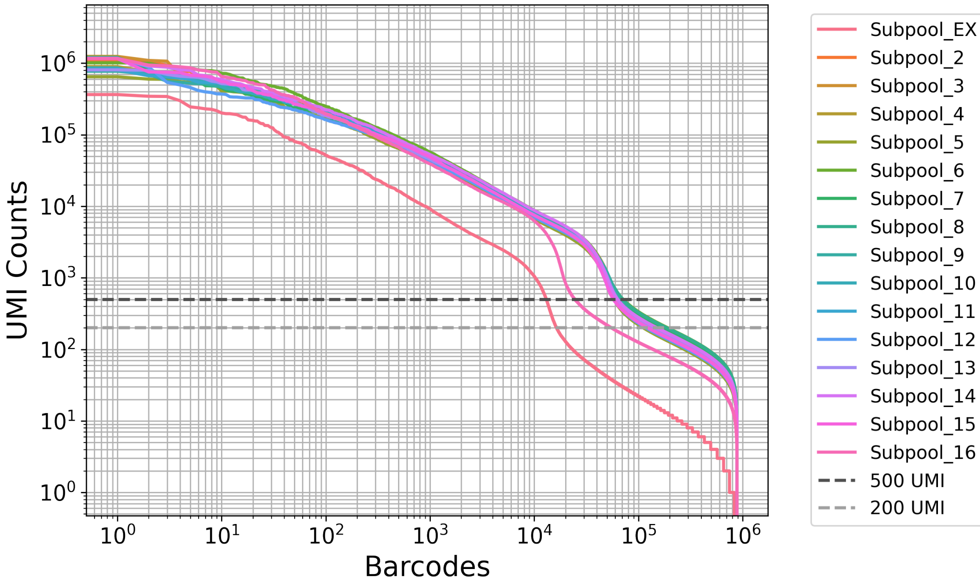


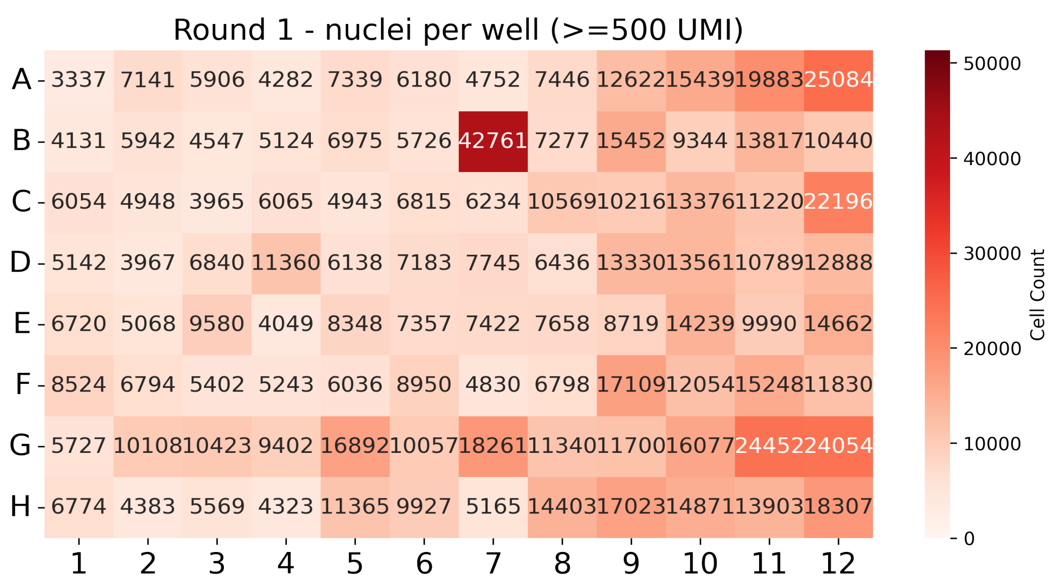
Table 3: QC stats for nuclei >=200 UMI

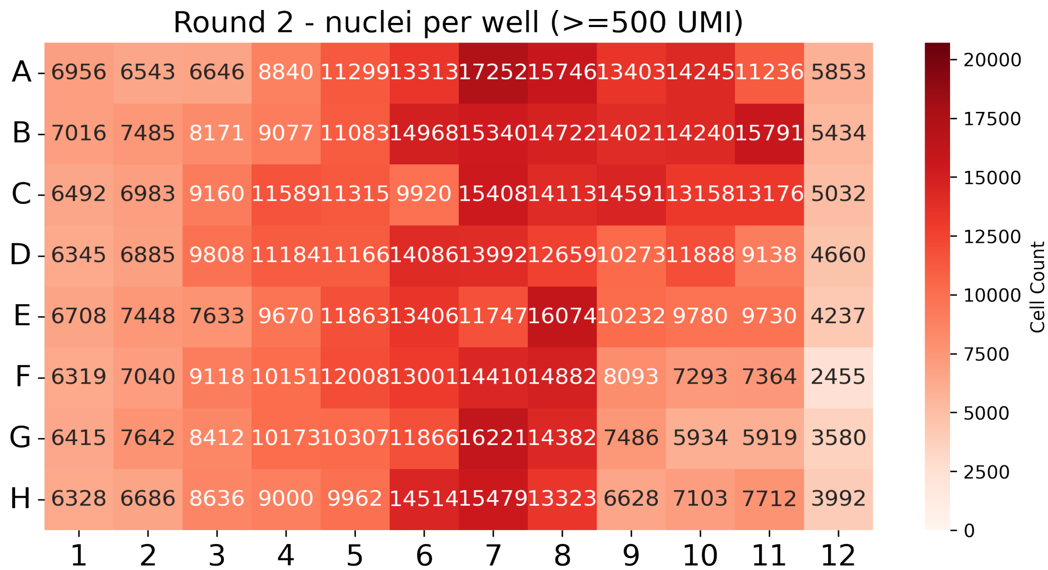
Subpool	Number of nuclei	Median UMIs	Mean UMIs	Median genes	Mean genes
Subpool_EX	16,416	1,463	3,453.9	1,006	1,418.2
Subpool_2	139,879	424	3,237.5	349	1,114.4
Subpool_3	131,911	451	3,201.3	367	1,112.8
Subpool_4	130,136	444	3,367.4	361	1,140.7
Subpool_5	112,100	569	3,157.0	449	1,147.4
Subpool_6	191,164	350	2,644.0	291	898.5
Subpool_7	170,192	354	2,511.4	295	917.4
Subpool_8	187,248	350	2,294.8	293	862.2
Subpool_9	145,873	414	2,896.9	343	1,046.3
Subpool_10	142,293	399	3,000.6	329	1,050.7
Subpool_11	128,023	452	3,231.9	368	1,124.5
Subpool_12	118,312	526	3,186.2	421	1,157.0
Subpool_13	143,504	401	2,877.5	330	1,031.9
Subpool_14	151,074	376	3,120.5	313	1,050.5
Subpool_15	123,542	435	3,247.3	357	1,114.8
Subpool_16	55,220	414	4,866.2	330	1,276.0

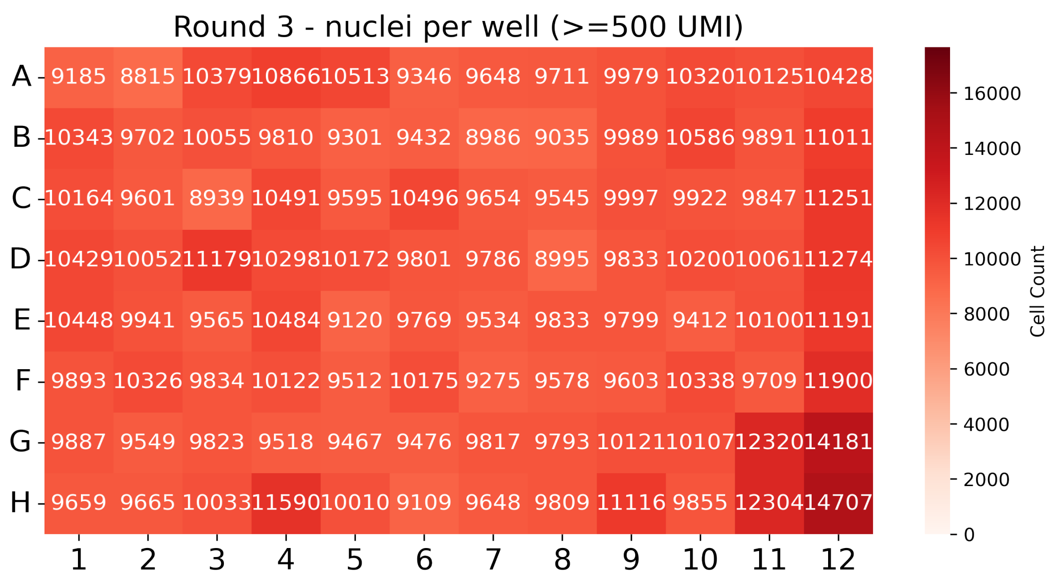
Table 4: QC stats for nuclei >=500 UMI

Subpool	Number of nuclei	Median UMIs	Mean UMIs	Median genes	Mean genes
Subpool_EX	13,092	1,912	4,251.7	1,249	1,711.2
Subpool_2	64,620	3,082	6,676.2	1,670	2,131.7
Subpool_3	62,953	2,923	6,395.0	1,611	2,067.4
Subpool_4	61,651	3,098	6,791.4	1,670	2,141.1
Subpool_5	58,391	2,759	5,798.8	1,535	1,982.6
Subpool_6	71,077	2,725	6,625.5	1,512	2,008.9
Subpool_7	65,810	2,758	6,042.3	1,534	1,990.7
Subpool_8	70,812	2,462	5,598.7	1,419	1,883.1
Subpool_9	66,229	2,799	6,038.0	1,566	2,014.1
Subpool_10	62,666	2,982	6,451.3	1,621	2,081.0
Subpool_11	61,318	3,006	6,438.0	1,638	2,086.1
Subpool_12	60,184	2,894	5,989.9	1,596	2,043.7
Subpool_13	63,347	2,907	6,157.4	1,596	2,033.0
Subpool_14	62,800	3,261	7,105.9	1,737	2,187.8
Subpool_15	57,622	3,044	6,634.4	1,652	2,113.0
Subpool_16	24,582	4,655	10,568.1	2,125	2,569.3

In this combinatorial barcoding assay, nuclei are barcoded in 96-well plates over three rounds. The first round assigns sample-specific barcodes, while the next two are applied to the pooled mix. Even distribution in the first round is important to ensure consistent sample representation.







4. Background removal

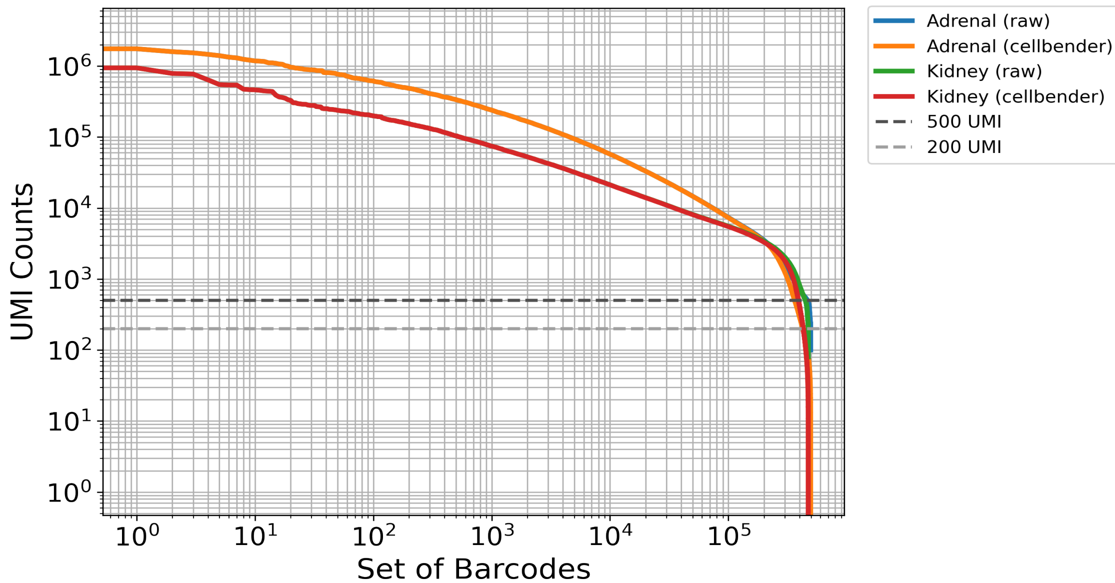
Table 5: Cellbender parameters

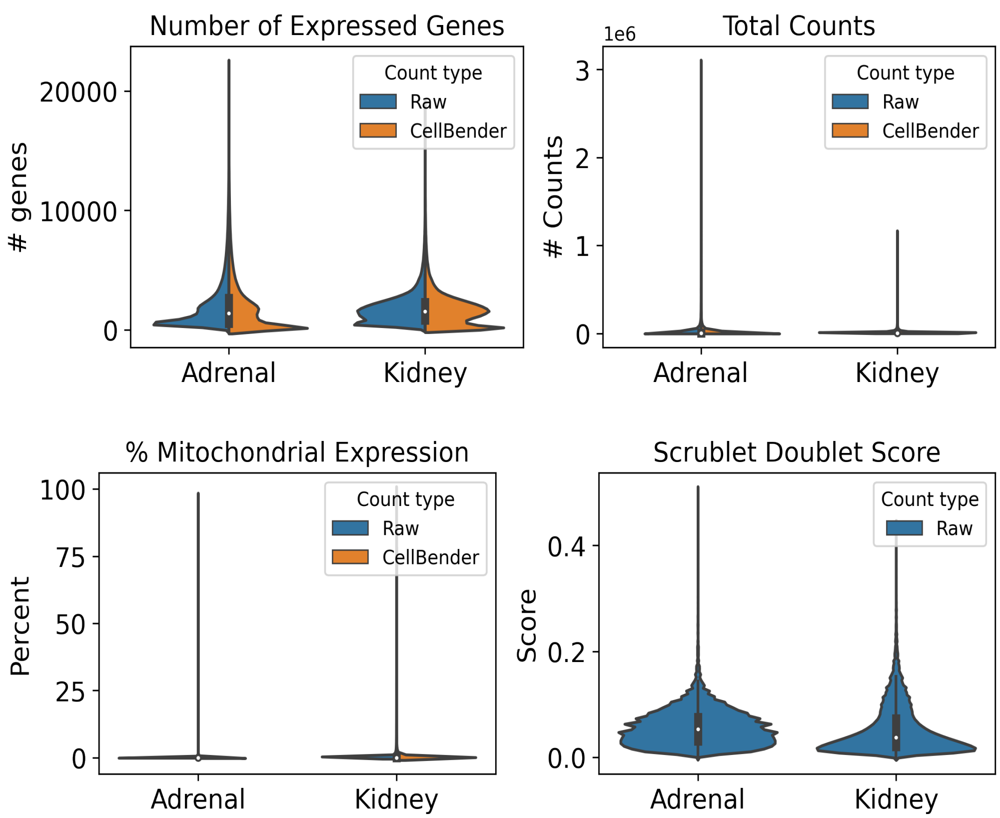
Subpool	droplets_included	learning_rate	expected_cells
Subpool_EX	40000	5e-05	13000
Subpool_2	200000	5e-05	67000
Subpool_3	200000	5e-05	67000
Subpool_4	200000	5e-05	67000
Subpool_5	200000	5e-05	67000

Subpool_6	200000	5e-05	67000
Subpool_7	200000	5e-05	67000
Subpool_8	200000	5e-05	67000
Subpool_9	200000	5e-05	67000
Subpool_10	200000	5e-05	67000
Subpool_11	200000	5e-05	67000
Subpool_12	200000	5e-05	67000
Subpool_13	200000	5e-05	67000
Subpool_14	200000	5e-05	67000
Subpool_15	200000	5e-05	67000
Subpool_16	200000	5e-05	67000

Table 6: Cellbender stats

Subpool	Percent counts removed	Found nuclei	Mean denoised counts
Subpool_EX	2.3	14,493	3,790.6
Subpool_2	3.4	67,028	6,226.6
Subpool_3	3.3	63,945	6,092.9
Subpool_4	3.4	66,787	6,089.7
Subpool_5	3.1	60,739	5,417.4
Subpool_6	3.9	72,345	6,260.9
Subpool_7	3.9	65,655	5,819.6
Subpool_8	4.3	70,939	5,346.7
Subpool_9	3.3	62,356	6,104.6
Subpool_10	3.4	62,263	6,269.0
Subpool_11	3.0	62,080	6,173.0
Subpool_12	3.2	59,666	5,842.6
Subpool_13	3.2	64,322	5,874.0
Subpool_14	3.5	69,770	6,212.4
Subpool_15	3.2	60,296	6,158.1
Subpool_16	2.3	47,379	5,505.4





5. Barcode sample map

The first round corresponds to sample barcoding. In Parse Bio assays, two barcodes are included in the first round: a random hexamer (R) and oligo-dT (T) barcode per well (parse barcode type). This reduces 3-prime bias and allows for the capture of full-length transcripts. These sample level barcodes can be found in the barcode-1 region in the seqspec, and begin at position 16 (0-based) in the full 24-nt cell barcode. For the WT_mega kit, samples are distributed across 96 wells in round 1. The following table includes our lab sample ID as sample descriptions and IGVF sample accessions.

Table 7: Barcode sample map

barcode	sample accession	sample description	position	parse barcode type	well
CATCATCC	IGVFSM9731QOVH	016_B6J_10F_08	16	R	A1
CATTCCTA	IGVFSM9731QOVH	016_B6J_10F_08	16	T	A1
CTGCTTTG	IGVFSM7981FEKQ	017_B6J_10M_08	16	R	A2
CTTCATCA	IGVFSM7981FEKQ	017_B6J_10M_08	16	T	A2
CTAAGGGA	IGVFSM7273NNBM	018_B6J_10F_08	16	R	A3
CCTATATC	IGVFSM7273NNBM	018_B6J_10F_08	16	T	A3
GCTTATAG	IGVFSM2795NYTC	019_B6J_10M_08	16	R	A4
ACATTTAC	IGVFSM2795NYTC	019_B6J_10M_08	16	T	A4
TCTGATCC	IGVFSM2926ETSH	020_B6J_10F_08	16	R	A5
ACTTAGCT	IGVFSM2926ETSH	020_B6J_10F_08	16	T	A5
TCTCTTGG	IGVFSM4409GRZD	021_B6J_10M_08	16	R	A6
CCAATTCT	IGVFSM4409GRZD	021_B6J_10M_08	16	T	A6
CAATTTCC	IGVFSM6796IKTY	024_B6J_10F_08	16	R	A7
GCCTATCT	IGVFSM6796IKTY	024_B6J_10F_08	16	T	A7
AGTCTCTT	IGVFSM4127LBLM	025_B6J_10M_08	16	R	A8
ATGCTGCT	IGVFSM4127LBLM	025_B6J_10M_08	16	T	A8
TGCTGCTC	IGVFSM0349VWWY	016_B6J_10F_09	16	R	A9
CATTTACA	IGVFSM0349VWWY	016_B6J_10F_09	16	T	A9
TGCTGCTC	IGVFSM7878IZMN	066_NODJ_10F_09	16	R	A9
CATTTACA	IGVFSM7878IZMN	066_NODJ_10F_09	16	T	A9
GTATTTCC	IGVFSM3241ASWH	017_B6J_10M_09	16	R	A10
ACTCGTAA	IGVFSM3241ASWH	017_B6J_10M_09	16	T	A10
GTATTTCC	IGVFSM4726ZWYC	067_NODJ_10M_09	16	R	A10
ACTCGTAA	IGVFSM4726ZWYC	067_NODJ_10M_09	16	T	A10
TTCCTGTG	IGVFSM9474HYEG	018_B6J_10F_09	16	R	A11
CCTTTGCA	IGVFSM9474HYEG	018_B6J_10F_09	16	T	A11
TTCCTGTG	IGVFSM4214PFWU	068_NODJ_10F_09	16	R	A11

CCTTTGCA	IGVFSM4214PFWU	068_NODJ_10F_09	16	T	A11
GCTGCTTC	IGVFSM9788LCSW	019_B6J_10M_09	16	R	A12
ACTCCTGC	IGVFSM9788LCSW	019_B6J_10M_09	16	T	A12
GCTGCTTC	IGVFSM8898IMBT	069_NODJ_10M_09	16	R	A12
ACTCCTGC	IGVFSM8898IMBT	069_NODJ_10M_09	16	T	A12
TATGTGTC	IGVFSM1981FBYZ	066_NODJ_10F_08	16	R	B1
ATTTGGCA	IGVFSM1981FBYZ	066_NODJ_10F_08	16	T	B1
CAATTCTC	IGVFSM0149TNQV	067_NODJ_10M_08	16	R	B2
TTATTCTG	IGVFSM0149TNQV	067_NODJ_10M_08	16	T	B2
TGGTCTCC	IGVFSM2286NVNW	068_NODJ_10F_08	16	R	B3
TCATGCTC	IGVFSM2286NVNW	068_NODJ_10F_08	16	T	B3
GCTCTTTA	IGVFSM0256YAXR	069_NODJ_10M_08	16	R	B4
CATACTTC	IGVFSM0256YAXR	069_NODJ_10M_08	16	T	B4
GCTGCATG	IGVFSM6603NGXT	070_NODJ_10F_08	16	R	B5
CCGTTCTA	IGVFSM6603NGXT	070_NODJ_10F_08	16	T	B5
ACTCATTT	IGVFSM2251OZKQ	071_NODJ_10M_08	16	R	B6
GCTTCATA	IGVFSM2251OZKQ	071_NODJ_10M_08	16	T	B6
AGTCTTGG	IGVFSM0170ECBA	072_NODJ_10F_08	16	R	B7
CTCTGTGC	IGVFSM0170ECBA	072_NODJ_10F_08	16	T	B7
GGTTCTTC	IGVFSM5573LEDH	073_NODJ_10M_08	16	R	B8
CCCTTATA	IGVFSM5573LEDH	073_NODJ_10M_08	16	T	B8
TCATGTTG	IGVFSM4800CIZG	020_B6J_10F_09	16	R	B9
ACTGCTCT	IGVFSM4800CIZG	020_B6J_10F_09	16	T	B9
TCATGTTG	IGVFSM7567RTBV	070_NODJ_10F_09	16	R	B9
ACTGCTCT	IGVFSM7567RTBV	070_NODJ_10F_09	16	T	B9
ATTTTGCC	IGVFSM7243TNKF	021_B6J_10M_09	16	R	B10
CTCTAATC	IGVFSM7243TNKF	021_B6J_10M_09	16	T	B10
ATTTTGCC	IGVFSM4531FRYY	071_NODJ_10M_09	16	R	B10
CTCTAATC	IGVFSM4531FRYY	071_NODJ_10M_09	16	T	B10
CTTCTGTA	IGVFSM8674PTVY	024_B6J_10F_09	16	R	B11
ACCCTTGC	IGVFSM8674PTVY	024_B6J_10F_09	16	T	B11
CTTCTGTA	IGVFSM4518CECN	072_NODJ_10F_09	16	R	B11
ACCCTTGC	IGVFSM4518CECN	072_NODJ_10F_09	16	T	B11
GTCCATCT	IGVFSM3060AXDG	025_B6J_10M_09	16	R	B12
ATCTTAGG	IGVFSM3060AXDG	025_B6J_10M_09	16	T	B12

GTCCATCT	IGVFSM4828VZOR	073_NODJ_10M_09	16	R	B12
ATCTTAGG	IGVFSM4828VZOR	073_NODJ_10M_09	16	T	B12
GCTATCTC	IGVFSM5489EQDV	026_AJ_10F_08	16	R	C1
CATGTCTC	IGVFSM5489EQDV	026_AJ_10F_08	16	T	C1
TAGTTTCC	IGVFSM9075IJZC	029_AJ_10M_08	16	R	C2
TCATTGCA	IGVFSM9075IJZC	029_AJ_10M_08	16	T	C2
TCCATTAT	IGVFSM6339UEAO	028_AJ_10F_08	16	R	C3
ACACCTTT	IGVFSM6339UEAO	028_AJ_10F_08	16	T	C3
AGGATTAA	IGVFSM4824RRCE	031_AJ_10M_08	16	R	C4
AATTTCTC	IGVFSM4824RRCE	031_AJ_10M_08	16	T	C4
AATCTTTC	IGVFSM7056LBHR	030_AJ_10F_08	16	R	C5
ATTCATGG	IGVFSM7056LBHR	030_AJ_10F_08	16	T	C5
GTCATATG	IGVFSM6013ZWZO	033_AJ_10M_08	16	R	C6
ACTTTACC	IGVFSM6013ZWZO	033_AJ_10M_08	16	T	C6
GTGCTTCC	IGVFSM8570TOBS	032_AJ_10F_08	16	R	C7
CTTCTAAC	IGVFSM8570TOBS	032_AJ_10F_08	16	T	C7
ATGTGTTG	IGVFSM4324XCGU	035_AJ_10M_08	16	R	C8
CTATTTCA	IGVFSM4324XCGU	035_AJ_10M_08	16	T	C8
CCATCTTG	IGVFSM3057IFAI	026_AJ_10F_09	16	R	C9
TCTCATGC	IGVFSM3057IFAI	026_AJ_10F_09	16	T	C9
CCATCTTG	IGVFSM9088FTIY	076_PWKJ_10F_09	16	R	C9
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TACTGTCT	IGVFSM8586FNQO	029_AJ_10M_09	16	R	C10
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ATCCTTAC	IGVFSM2207LORL	077_PWKJ_10M_09	16	T	C10
TTCATCGC	IGVFSM3536NVMG	028_AJ_10F_09	16	R	C11
TAAATATC	IGVFSM3536NVMG	028_AJ_10F_09	16	T	C11
TTCATCGC	IGVFSM4211TCCR	078_PWKJ_10F_09	16	R	C11
TAAATATC	IGVFSM4211TCCR	078_PWKJ_10F_09	16	T	C11
ACTGTGGG	IGVFSM4523NJGP	031_AJ_10M_09	16	R	C12
TTACCTGC	IGVFSM4523NJGP	031_AJ_10M_09	16	T	C12
ACTGTGGG	IGVFSM4917CZZF	079_PWKJ_10M_09	16	R	C12
TTACCTGC	IGVFSM4917CZZF	079_PWKJ_10M_09	16	T	C12
TCTGTGCC	IGVFSM5120VDBH	076_PWKJ_10F_08	16	R	D1

CACTTTCA	IGVFSM5120VDBH	076_PWKJ_10F_08	16	T	D1
TCAATCTC	IGVFSM2744HFBR	077_PWKJ_10M_08	16	R	D2
CACCTTTA	IGVFSM2744HFBR	077_PWKJ_10M_08	16	T	D2
GTCCTCTG	IGVFSM9958NNHY	078_PWKJ_10F_08	16	R	D3
CTGACTTC	IGVFSM9958NNHY	078_PWKJ_10F_08	16	T	D3
TTACATTC	IGVFSM0701NELT	079_PWKJ_10M_08	16	R	D4
CATTTGGA	IGVFSM0701NELT	079_PWKJ_10M_08	16	T	D4
ATTCTGTC	IGVFSM1448KEUX	080_PWKJ_10F_08	16	R	D5
GCTCTACT	IGVFSM1448KEUX	080_PWKJ_10F_08	16	T	D5
TGTGTATG	IGVFSM8451AOOU	081_PWKJ_10M_08	16	R	D6
GTTACGTA	IGVFSM8451AOOU	081_PWKJ_10M_08	16	T	D6
TCCATTTG	IGVFSM4227UNYC	084_PWKJ_10F_08	16	R	D7
CCTGTTGC	IGVFSM4227UNYC	084_PWKJ_10F_08	16	T	D7
TTAGCTTC	IGVFSM1696GVKP	085_PWKJ_10M_08	16	R	D8
CTATCATC	IGVFSM1696GVKP	085_PWKJ_10M_08	16	T	D8
GTGCTTGA	IGVFSM7352LZKM	030_AJ_10F_09	16	R	D9
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GTGCTTGA	IGVFSM4983BATQ	080_PWKJ_10F_09	16	R	D9
GCTATCAT	IGVFSM4983BATQ	080_PWKJ_10F_09	16	T	D9
GTTTGTGA	IGVFSM6020DPIE	033_AJ_10M_09	16	R	D10
ACATTCAT	IGVFSM6020DPIE	033_AJ_10M_09	16	T	D10
GTTTGTGA	IGVFSM8581GLVP	081_PWKJ_10M_09	16	R	D10
ACATTCAT	IGVFSM8581GLVP	081_PWKJ_10M_09	16	T	D10
GAAATTAG	IGVFSM3951CGSL	032_AJ_10F_09	16	R	D11
TTCGCTAC	IGVFSM3951CGSL	032_AJ_10F_09	16	T	D11
GAAATTAG	IGVFSM3357EWRJ	084_PWKJ_10F_09	16	R	D11
TTCGCTAC	IGVFSM3357EWRJ	084_PWKJ_10F_09	16	T	D11
GCAAATTC	IGVFSM3981QKEE	035_AJ_10M_09	16	R	D12
CATTCTAC	IGVFSM3981QKEE	035_AJ_10M_09	16	T	D12
GCAAATTC	IGVFSM8335WDIG	085_PWKJ_10M_09	16	R	D12
CATTCTAC	IGVFSM8335WDIG	085_PWKJ_10M_09	16	T	D12
GAGGTTGA	IGVFSM1632QLUR	036_129S1J_10F_08	16	R	E1
CACTTATC	IGVFSM1632QLUR	036_129S1J_10F_08	16	T	E1
CCTGTCTG	IGVFSM9668NECE	037_129S1J_10M_08	16	R	E2
ATAAGCTC	IGVFSM9668NECE	037_129S1J_10M_08	16	T	E2

GTGGGTTC	IGVFSM0340EAYH	038_129S1J_10F_08	16	R	E3
TCATCCTG	IGVFSM0340EAYH	038_129S1J_10F_08	16	T	E3
TTTGCATC	IGVFSM4755QPGT	041_129S1J_10M_08	16	R	E4
CCTGGTAT	IGVFSM4755QPGT	041_129S1J_10M_08	16	T	E4
AGGTAATA	IGVFSM9964SESF	040_129S1J_10F_08	16	R	E5
TGGTATAC	IGVFSM9964SESF	040_129S1J_10F_08	16	T	E5
GTGCCTTC	IGVFSM4576HVJC	043_129S1J_10M_08	16	R	E6
TTGGGAGA	IGVFSM4576HVJC	043_129S1J_10M_08	16	T	E6
ATGTTTCC	IGVFSM3033SSIU	044_129S1J_10F_08	16	R	E7
ACTTCATC	IGVFSM3033SSIU	044_129S1J_10F_08	16	T	E7
CTTAATTC	IGVFSM3513BNRI	045_129S1J_10M_08	16	R	E8
TCTCTAGC	IGVFSM3513BNRI	045_129S1J_10M_08	16	T	E8
TCTGGCTC	IGVFSM7015STPL	036_129S1J_10F_09	16	R	E9
ATGCCCTT	IGVFSM7015STPL	036_129S1J_10F_09	16	T	E9
TCTGGCTC	IGVFSM7915MIHZ	086_CASTJ_10F_09	16	R	E9
ATGCCCTT	IGVFSM7915MIHZ	086_CASTJ_10F_09	16	T	E9
CATCATTT	IGVFSM3870IKYE	037_129S1J_10M_09	16	R	E10
CCCAATTT	IGVFSM3870IKYE	037_129S1J_10M_09	16	T	E10
CATCATTT	IGVFSM1924DVVF	087_CASTJ_10M_09	16	R	E10
CCCAATTT	IGVFSM1924DVVF	087_CASTJ_10M_09	16	T	E10
GTTGTCTC	IGVFSM2985VTWQ	038_129S1J_10F_09	16	R	E11
ACTATATA	IGVFSM2985VTWQ	038_129S1J_10F_09	16	T	E11
GTTGTCTC	IGVFSM8639WEBM	088_CASTJ_10F_09	16	R	E11
ACTATATA	IGVFSM8639WEBM	088_CASTJ_10F_09	16	T	E11
ATCTTCTG	IGVFSM1728JVMJ	041_129S1J_10M_09	16	R	E12
CTCTATAC	IGVFSM1728JVMJ	041_129S1J_10M_09	16	T	E12
ATCTTCTG	IGVFSM0501BSTV	089_CASTJ_10M_09	16	R	E12
CTCTATAC	IGVFSM0501BSTV	089_CASTJ_10M_09	16	T	E12
TGTTTGCC	IGVFSM2352QEZL	086_CASTJ_10F_08	16	R	F1
CTGTCTCA	IGVFSM2352QEZL	086_CASTJ_10F_08	16	T	F1
TTCTGTCA	IGVFSM4914MZNP	087_CASTJ_10M_08	16	R	F2
GACCTTTC	IGVFSM4914MZNP	087_CASTJ_10M_08	16	T	F2
ACGGACTC	IGVFSM2528JPFP	088_CASTJ_10F_08	16	R	F3
GATTTGGC	IGVFSM2528JPFP	088_CASTJ_10F_08	16	T	F3
TTTGGTCA	IGVFSM0740TEWR	089_CASTJ_10M_08	16	R	F4

CGTCTAGG	IGVFSM0740TEWR	089_CASTJ_10M_08	16	T	F4
TATCCGGG	IGVFSM8717VBUZ	090_CASTJ_10F_08	16	R	F5
TACTCGAA	IGVFSM8717VBUZ	090_CASTJ_10F_08	16	T	F5
TGTCATTC	IGVFSM5442GAOY	091_CASTJ_10M_08	16	R	F6
CAGCCTTT	IGVFSM5442GAOY	091_CASTJ_10M_08	16	T	F6
ATTCTCTG	IGVFSM4807XVYB	094_CASTJ_10F_08	16	R	F7
CCTCATTA	IGVFSM4807XVYB	094_CASTJ_10F_08	16	T	F7
TGGCTTCC	IGVFSM5275FTRH	095_CASTJ_10M_08	16	R	F8
CTTATACC	IGVFSM5275FTRH	095_CASTJ_10M_08	16	T	F8
TTGTTGCC	IGVFSM7656VCGX	040_129S1J_10F_09	16	R	F9
TCTATTAC	IGVFSM7656VCGX	040_129S1J_10F_09	16	T	F9
TTGTTGCC	IGVFSM6781WLWW	090_CASTJ_10F_09	16	R	F9
TCTATTAC	IGVFSM6781WLWW	090_CASTJ_10F_09	16	T	F9
GTCATCTC	IGVFSM3274UVBQ	043_129S1J_10M_09	16	R	F10
CCTGCATT	IGVFSM3274UVBQ	043_129S1J_10M_09	16	T	F10
GTCATCTC	IGVFSM7315OJAM	091_CASTJ_10M_09	16	R	F10
CCTGCATT	IGVFSM7315OJAM	091_CASTJ_10M_09	16	T	F10
TTGCTCAT	IGVFSM9083BJCW	044_129S1J_10F_09	16	R	F11
CAATCCTT	IGVFSM9083BJCW	044_129S1J_10F_09	16	T	F11
TTGCTCAT	IGVFSM3937HVNK	094_CASTJ_10F_09	16	R	F11
CAATCCTT	IGVFSM3937HVNK	094_CASTJ_10F_09	16	T	F11
CTGTCTGC	IGVFSM2309TUXE	045_129S1J_10M_09	16	R	F12
TTGTCTTA	IGVFSM2309TUXE	045_129S1J_10M_09	16	T	F12
CTGTCTGC	IGVFSM0131ZKOP	093_CASTJ_10M_09	16	R	F12
TTGTCTTA	IGVFSM0131ZKOP	093_CASTJ_10M_09	16	T	F12
TATATTCC	IGVFSM0232VKZB	058_WSBJ_10F_08	16	R	G1
TCACTTTA	IGVFSM0232VKZB	058_WSBJ_10F_08	16	T	G1
ATATTGGC	IGVFSM9417BLQR	057_WSBJ_10M_08	16	R	G2
TGCTTGGG	IGVFSM9417BLQR	057_WSBJ_10M_08	16	T	G2
GTGTCCTC	IGVFSM0760BMNA	060_WSBJ_10F_08	16	R	G3
CGCTCATT	IGVFSM0760BMNA	060_WSBJ_10F_08	16	T	G3
ATCTTCAT	IGVFSM4075DZPE	059_WSBJ_10M_08	16	R	G4
GCCTCTAT	IGVFSM4075DZPE	059_WSBJ_10M_08	16	T	G4
CGTGGTTG	IGVFSM4265ZXJM	062_WSBJ_10F_08	16	R	G5
GAGCACAA	IGVFSM4265ZXJM	062_WSBJ_10F_08	16	T	G5

TTGCATCC	IGVFSM8686RRRP	061_WSBJ_10M_08	16	R	G6
CTCTTAAC	IGVFSM8686RRRP	061_WSBJ_10M_08	16	T	G6
TCTTAATC	IGVFSM3681WOHF	096_WSBJ_10F_08	16	R	G7
TCTAGGCT	IGVFSM3681WOHF	096_WSBJ_10F_08	16	T	G7
TGCATTTC	IGVFSM7664LZLU	063_WSBJ_10M_08	16	R	G8
AATTCTGC	IGVFSM7664LZLU	063_WSBJ_10M_08	16	T	G8
GATGTTTC	IGVFSM2869NKDL	046_NZOJ_10F_09	16	R	G9
CATTCTCA	IGVFSM2869NKDL	046_NZOJ_10F_09	16	T	G9
GATGTTTC	IGVFSM9077YRZW	058_WSBJ_10F_09	16	R	G9
CATTCTCA	IGVFSM9077YRZW	058_WSBJ_10F_09	16	T	G9
ATCTTGTC	IGVFSM8082YJJT	047_NZOJ_10M_09	16	R	G10
ACTTGCCT	IGVFSM8082YJJT	047_NZOJ_10M_09	16	T	G10
ATCTTGTC	IGVFSM1334FZTF	057_WSBJ_10M_09	16	R	G10
ACTTGCCT	IGVFSM1334FZTF	057_WSBJ_10M_09	16	T	G10
TCATATTC	IGVFSM4505WWVP	048_NZOJ_10F_09	16	R	G11
ATCATTGC	IGVFSM4505WWVP	048_NZOJ_10F_09	16	T	G11
TCATATTC	IGVFSM6194HVVO	060_WSBJ_10F_09	16	R	G11
ATCATTGC	IGVFSM6194HVVO	060_WSBJ_10F_09	16	T	G11
TGGCCTCT	IGVFSM7894QTQT	049_NZOJ_10M_09	16	R	G12
GTTCAACA	IGVFSM7894QTQT	049_NZOJ_10M_09	16	T	G12
TGGCCTCT	IGVFSM7285SRRG	059_WSBJ_10M_09	16	R	G12
GTTCAACA	IGVFSM7285SRRG	059_WSBJ_10M_09	16	T	G12
CGTTGTCT	IGVFSM9456XQYT	046_NZOJ_10F_08	16	R	H1
CCATTTGC	IGVFSM9456XQYT	046_NZOJ_10F_08	16	T	H1
TCTTGTCA	IGVFSM8242IPDC	047_NZOJ_10M_08	16	R	H2
GACTTTGC	IGVFSM8242IPDC	047_NZOJ_10M_08	16	T	H2
TATTCCTG	IGVFSM4848LHPG	048_NZOJ_10F_08	16	R	H3
ATTGGCTC	IGVFSM4848LHPG	048_NZOJ_10F_08	16	T	H3
TCCATGTC	IGVFSM3199ZHGC	049_NZOJ_10M_08	16	R	H4
GTGCTAGC	IGVFSM3199ZHGC	049_NZOJ_10M_08	16	T	H4
TTGTCATC	IGVFSM4326PPHF	050_NZOJ_10F_08	16	R	H5
CTTTCAAC	IGVFSM4326PPHF	050_NZOJ_10F_08	16	T	H5
ATTTCCTG	IGVFSM7580WBSP	051_NZOJ_10M_08	16	R	H6
ACTATTGC	IGVFSM7580WBSP	051_NZOJ_10M_08	16	T	H6
GTGTCTCC	IGVFSM0525KCUV	052_NZOJ_10F_08	16	R	H7

ACTGGCTT	IGVFSM0525KCUV	052_NZOJ_10F_08	16	T	H7
GTGTGTGT	IGVFSM5808CHQB	053_NZOJ_10M_08	16	R	H8
ATTAGGCT	IGVFSM5808CHQB	053_NZOJ_10M_08	16	T	H8
TATGCTTC	IGVFSM9303HUHF	050_NZOJ_10F_09	16	R	H9
GCCTTTCA	IGVFSM9303HUHF	050_NZOJ_10F_09	16	T	H9
TATGCTTC	IGVFSM2970OEMT	062_WSBJ_10F_09	16	R	H9
GCCTTTCA	IGVFSM2970OEMT	062_WSBJ_10F_09	16	T	H9
ATGGTGTT	IGVFSM0181XWPQ	051_NZOJ_10M_09	16	R	H10
ATTCTAGG	IGVFSM0181XWPQ	051_NZOJ_10M_09	16	T	H10
ATGGTGTT	IGVFSM0104SYAP	061_WSBJ_10M_09	16	R	H10
ATTCTAGG	IGVFSM0104SYAP	061_WSBJ_10M_09	16	T	H10
GAATAATG	IGVFSM1587OINL	052_NZOJ_10F_09	16	R	H11
CCTTACAT	IGVFSM1587OINL	052_NZOJ_10F_09	16	T	H11
GAATAATG	IGVFSM9060ZOSA	096_WSBJ_10F_09	16	R	H11
CCTTACAT	IGVFSM9060ZOSA	096_WSBJ_10F_09	16	T	H11
CCTCTGTG	IGVFSM5770UUHX	053_NZOJ_10M_09	16	R	H12
ACATTTGG	IGVFSM5770UUHX	053_NZOJ_10M_09	16	T	H12
CCTCTGTG	IGVFSM4351BICU	063_WSBJ_10M_09	16	R	H12
ACATTTGG	IGVFSM4351BICU	063_WSBJ_10M_09	16	T	H12

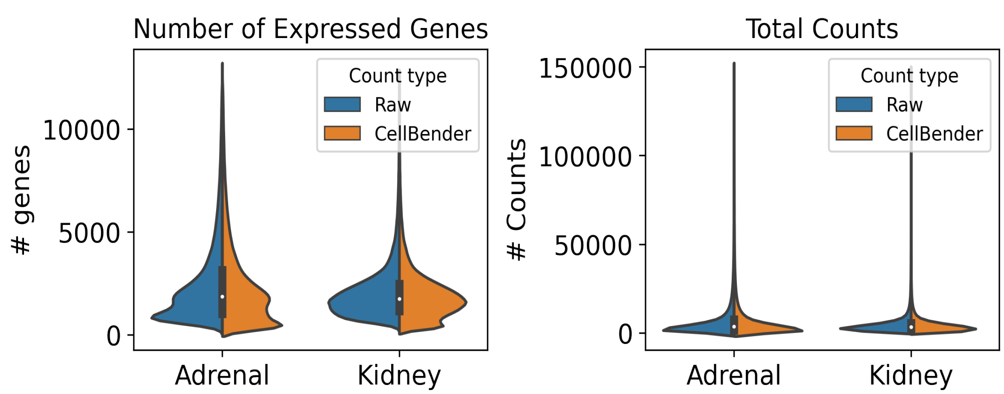
6. Description of data processing workflow and h5ad files

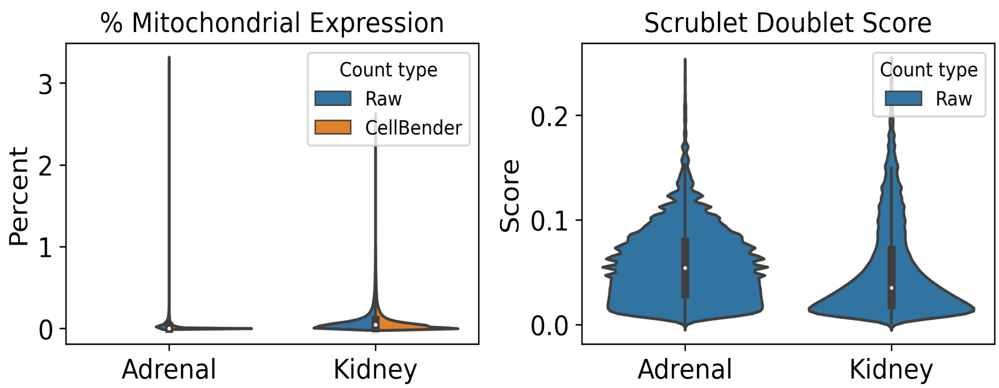
Data from all 16 subpools were concatenated into one AnnData object per tissue and combined with matching data from other plates for processing and cell type annotation. The AnnData contains detailed sample and cell metadata in the observation (obs) table and gene information in the variables table (var). Cell IDs (adata.obs.index) are re-formatted to be human-readable and unique across subpools and experiments by appending subpool and experiment IDs. The final AnnData contains both raw and CellBender denoised counts matrices in separate layers: adata.layers['raw_counts'] and adata.layers['cellbender_counts']. The current adata.X points to the CellBender denoised counts. Cells or nuclei are filtered by >0.5 cell_probability from CellBender analysis.

In addition to CellBender filtering, nuclei were also filtered by the following QC parameters:

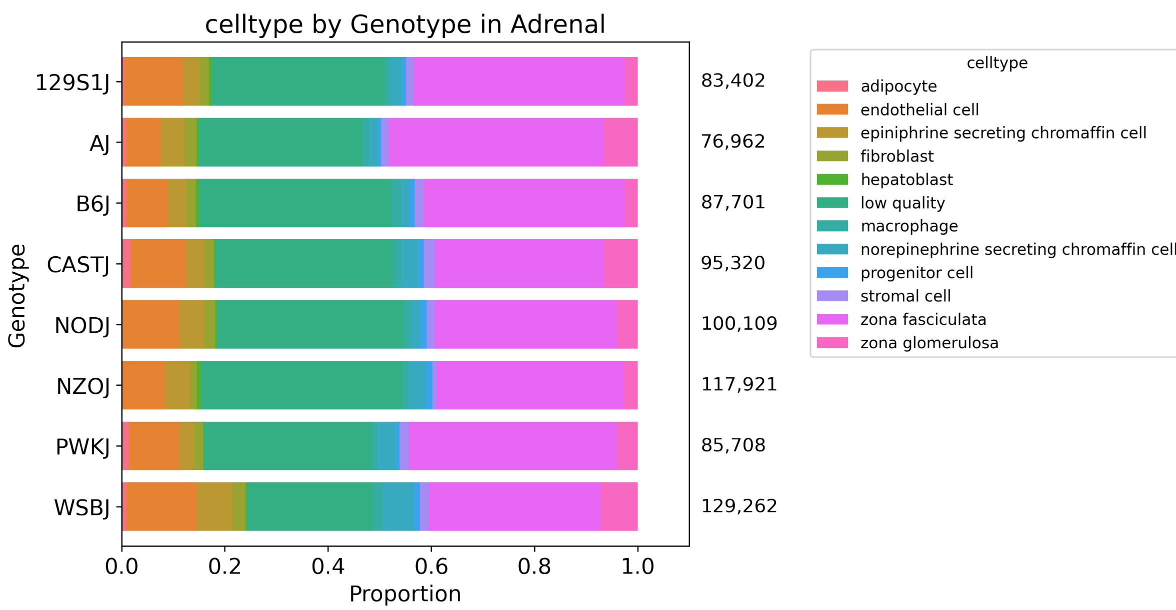
Table 8: QC thresholds

Parameter	Threshold
Minimum # UMIs	500.0
Maximum # UMIs	150000.0
Minimum # genes	250.0
Maximum % mito.	1.0
Maximum doublet score	0.25





After filtering, data for each tissue was normalized and clustered using CellBender denoised counts. Normalization includes regression of technical parameters (number of genes expressed per cell and percent mitochondrial gene expression). Harmony was used to integrate exome capture and non-exome capture subpools for annotation. A Leiden clustering resolution of 1 was used for 41 total clusters in Adrenal and 54 clusters in Kidney, using all tissue data across experiments. Cell type annotations were assigned per Leiden cluster or subcluster (leiden_R) using expression of canonical cell type marker genes.



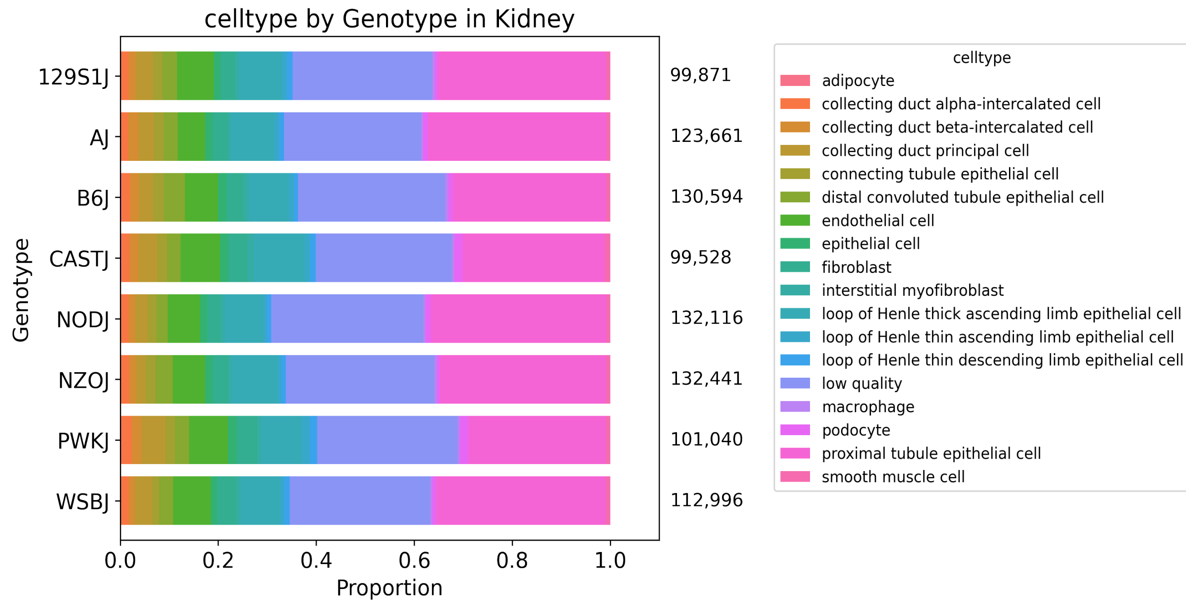


Table 9: Description of obs keys in h5ad file

Key	Description
cellID	Index of obs table. Includes barcoding wells for each round (1_2_3), subpool, and experiment.
lab_sample_id	Human-readable sample name used in-house.
sample	IGVF tissue accession.
plate	Experiment ID, e.g. igvf_003.
subpool	Subpool ID, e.g. Subpool_2. Number corresponds to Illumina index ID from Parse Bio kit.
SampleType	Nuclei or Cells.
Tissue	Human-readable tissue name, e.g. Kidney.
Sex	Male or Female.
Age	Age in postnatal months (PNM) of the mouse, e.g. PNM_02.
Genotype	Mouse genotype/strain, e.g. B6J.
leiden	Numeric Leiden cluster starting from 0.
leiden_R	If necessary, Leiden clusters and any sub-clusters, e.g. 15_1.
subpool_type	Exome capture (EX) or no exome capture (NO).
general_celltype	Broadest level of cell type annotations, e.g. neuron.
general_CL_ID	EMBL CL database ID for general cell type annotation, e.g. CL:0000540.
celltype	Finest level of cell type annotation with a CL ID, e.g. Vip+ GABAergic cortical interneuron.
CL_ID	EMBL CL database ID for cell type annotation, CL:4023016
subtype	Finest level of cell type annotation that may not have a CL ID. Mostly matches celltype.
Protocol	Cell barcoding and library building protocol/kit, e.g. Parse_WT.

Chemistry	Protocol chemistry version, v2 or v3.
bc	24-nucleotide sequence corresponding to 3 8-nucleotide barcode rounds (3, 2, 1).
bc1_sequence	8-nucleotide sequence barcode for round 1.
bc2_sequence	8-nucleotide sequence barcode for round 2.
bc3_sequence	8-nucleotide sequence barcode for round 3.
bc1_well	Round 1 well.
bc2_well	Round 2 well.
bc3_well	Round 2 well.
Row	Sample barcoding plate (round 1) row.
Column	Sample barcoding plate (round 1) column.
well_type	Single or Multiplexed.
Mouse_Tissue_ID	Same as lab_sample_id.
Multiplexed_sample1	Sample 1 from multiplexed well (if applicable)
Multiplexed_sample2	Sample 2 from multiplexed well (if applicable)
DOB	Date of birth of the mouse in month/day/year format.
Age_days	Age of the mouse in days.
Body_weight_g	Body weight in grams of the mouse.
Estrus_cycle	Estimated estrus stage of the mouse, if applicable.
Dissection_date	Date the tissue was harvested in month/day/year format.
Dissection_time	Time in hours:minutes (24-hour time) the mouse was sacrificed.
ZT	Number of hours into the light cycle, from lights on (06:30, ZT0) to lights off (20:30, ZT14).
Dissector	Initials of technician who dissected the tissue.
Tissue_weight_mg	Tissue weight in milligrams.
Notes	Notes taken during sample collection, experiment, or data processing.
n_genes_by_counts_raw	The number of genes with at least 1 count, calculated using raw counts.
total_counts_raw	Total counts (UMIs), calculated using raw counts.
total_counts_mt_raw	Total counts for mitochondrial genes, calculated using raw counts.
pct_counts_mt_raw	Percent of total counts in mitochondrial genes, calculated using raw counts.
n_genes_by_counts_cb	The number of genes with at least 1 count, calculated using CellBender counts.
total_counts_cb	Total counts (UMIs), calculated using CellBender counts.
total_counts_mt_cb	Total counts for mitochondrial genes, calculated using CellBender counts.
pct_counts_mt_cb	Percent of total counts in mitochondrial genes, calculated using CellBender counts.
doublet_score	Scrublet doublet score, calculated using CellBender counts.
predicted_doublet	Scrublet predicted doublet (True or False).
background_fraction	Fraction of counts CellBender predicted to be background noise.

cell_probability	CellBender probability that the cell is real.
cell_size	CellBender size factor.
droplet_efficiency	CellBender statistic indicating transcript capture efficiency per cell.
B6J_klue_counts	Estimated counts for B6J genotype from klue analysis.
NODJ_klue_counts	Estimated counts for NODJ genotype from klue analysis.
AJ_klue_counts	Estimated counts for AJ genotype from klue analysis.
PWKJ_klue_counts	Estimated counts for PWKJ genotype from klue analysis.
129S1J_klue_counts	Estimated counts for 129S1J genotype from klue analysis.
CASTJ_klue_counts	Estimated counts for CASTJ genotype from klue analysis.
WSBJ_klue_counts	Estimated counts for WSBJ genotype from klue analysis.
NZOJ_klue_counts	Estimated counts for NZOJ genotype from klue analysis.

7. Package versions

Table 10: Package versions

package	version
kallisto	0.50.1
bustools	0.43.2
kb	0.28.2
Python	3.9.0
snakemake	7.32.0
cellbender	0.3.2
scanpy	1.10.2
scrublet	0.2.3
anndata	0.10.8
pandas	2.2.2
numpy	1.26.4
harmonypy	0.0.9

8. Contact information and GitHub

Ali Mortazavi, ali.mortazavi@uci.edu

Elisabeth Rebboah, erebboah@uci.edu

Pipeline repo: https://github.com/mortazavilab/parse_pipeline

Code used to make this report: https://github.com/mortazavilab/8cube_manuscript/blob/main/plate_report/Plate_report_igvf_009.ipynb

Code used to process and annotate the data for main tissue: https://github.com/mortazavilab/8cube_manuscript/blob/main/annotation/Adrenal.ipynb

Code used to process and annotate the data for multiplexed tissue: https://github.com/mortazavilab/8cube_manuscript/blob/main/annotation/Kidney.ipynb